

Name: \_\_\_\_\_

**In the diagram,  $\triangle MNP \cong \triangle RST$ . Complete each statement.**

1.  $\overline{MN} \cong$  \_\_\_\_\_

2.  $\angle N \cong$  \_\_\_\_\_

3.  $\overline{PM} \cong$  \_\_\_\_\_

**In the diagram,  $\triangle ABC \cong \triangle DEF$ . Solve for the variable.**

4.  $AB = 2x + 1$  and  $DE = 15$ . Find  $x$ .

5.  $m\angle C = (4y - 6)^\circ$  and  $m\angle F = 46^\circ$ . Find  $y$ .

6.  $DE = 15$ ,  $EF = 12$ , and  $FD = 9$ . Find the perimeter of  $\triangle ABC$ .  
Explain how you know.

**Name the property of triangle congruence.**

7.  $\triangle GHJ \cong \triangle GHJ$

8. If  $\triangle GHJ \cong \triangle KLM$ , then  $\triangle KLM \cong \triangle GHJ$ .

9. If  $\triangle GHJ \cong \triangle KLM$  and  $\triangle KLM \cong \triangle PQR$ , then  $\triangle GHJ \cong \triangle PQR$ .

**Answer the question.**

10. A classmate writes  $\triangle ABC \cong \triangle EFD$  after being told  $\triangle ABC \cong \triangle DEF$ . Is the classmate correct? Explain.